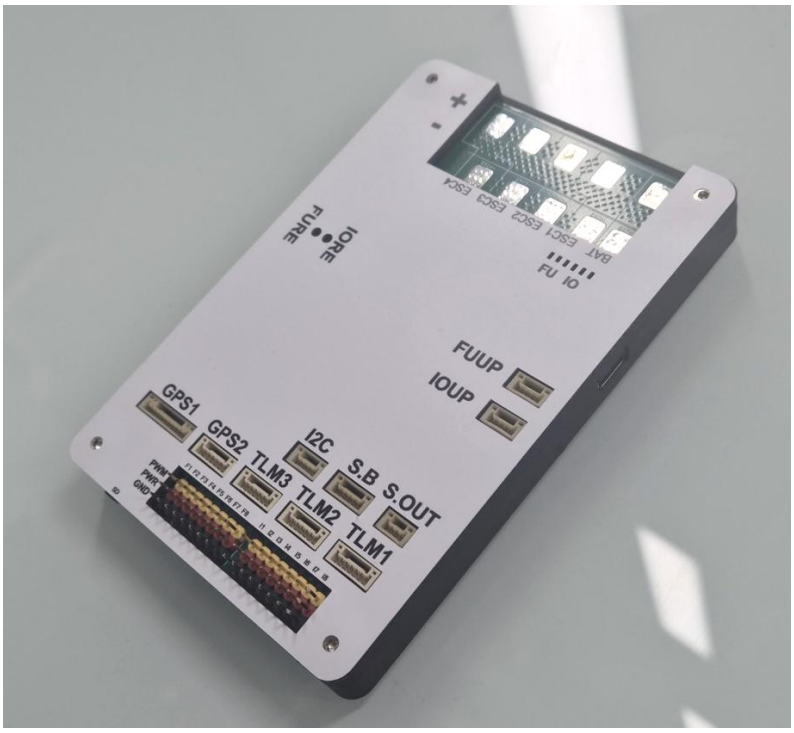
**Corp.DCLASS**

# DCD FC

Integrated power distribution flight controller | Product and connection guide

DCD FC is a flight controller with integrated power distribution developed by Corp.DCLASS. It combines battery input and power connections for four external ESCs on one board. GPS, telemetry, receiver interfaces and 16 PWM outputs support aircraft integration.




DCD FC with enclosure

## Specifications

| | |
|---|---|
| **Processors** | STM32H743IIT6 main MCU / STM32F103C8T6 IO MCU |
| **IMUs** | BMI088 + ICM-42688-P accelerometers and gyroscopes |
| **Barometer / compass** | MS5611 / IST8310 |
| **Storage** | FRAM parameters / microSD flight logs |
| **Outputs / interfaces** | 16 PWM outputs, 2 GPS, 3 telemetry, I2C, SBUS input, USB |
| **Input voltage** | Up to 25.2 V (6S LiPo) / supports 3–6S |
| **Power distribution** | Power pads for 4 external ESCs / ESCs not included |
| **Size / weight** | Approx. 126 × 84 × 13 mm / 103 g with enclosure |

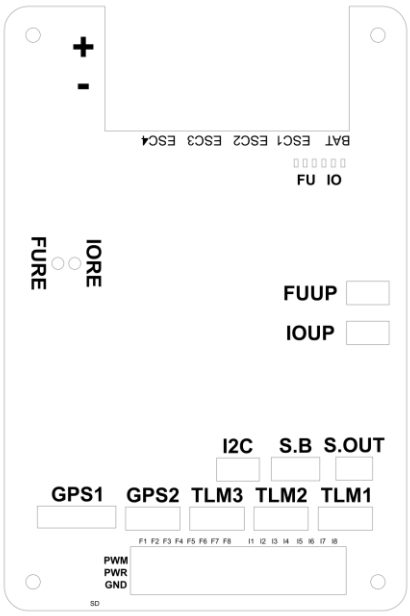
Dimensions refer to the enclosure, excluding protruding connectors and pins.

Purchase and technical support | Corp.DCLASS | https://www.d-class.co.kr

Connect to a PC using a standard USB data cable. Flight logs are stored on microSD.

## Ports and connections

Use the top-panel labels and the table below to identify ports and their functions.




| | |
|---|---|
| **BAT / ESC1~4** | Battery input / external ESC power |
| **GPS1 / GPS2** | GPS / SERIAL3 / SERIAL4 |
| **TLM1 / TLM2** | Telemetry / SERIAL1 / SERIAL2 |
| **TLM3** | Additional UART / SERIAL5 |
| **I2C** | External compass and I2C peripherals |
| **S.B** | SBUS receiver input |
| **F1~F8** | FMU PWM / SERVO9–16 |
| **I1~I8** | IO PWM / SERVO1–8 |
| **USB / SD** | PC connection / microSD logs |
| **FUUP / IOUP** | FMU / IO programming via SWD |

S.OUT is the IO SBUS output. See page 8 for FURE/IORE reset buttons and FU/IO LEDs. PPM input and external CAN ports are not supported.

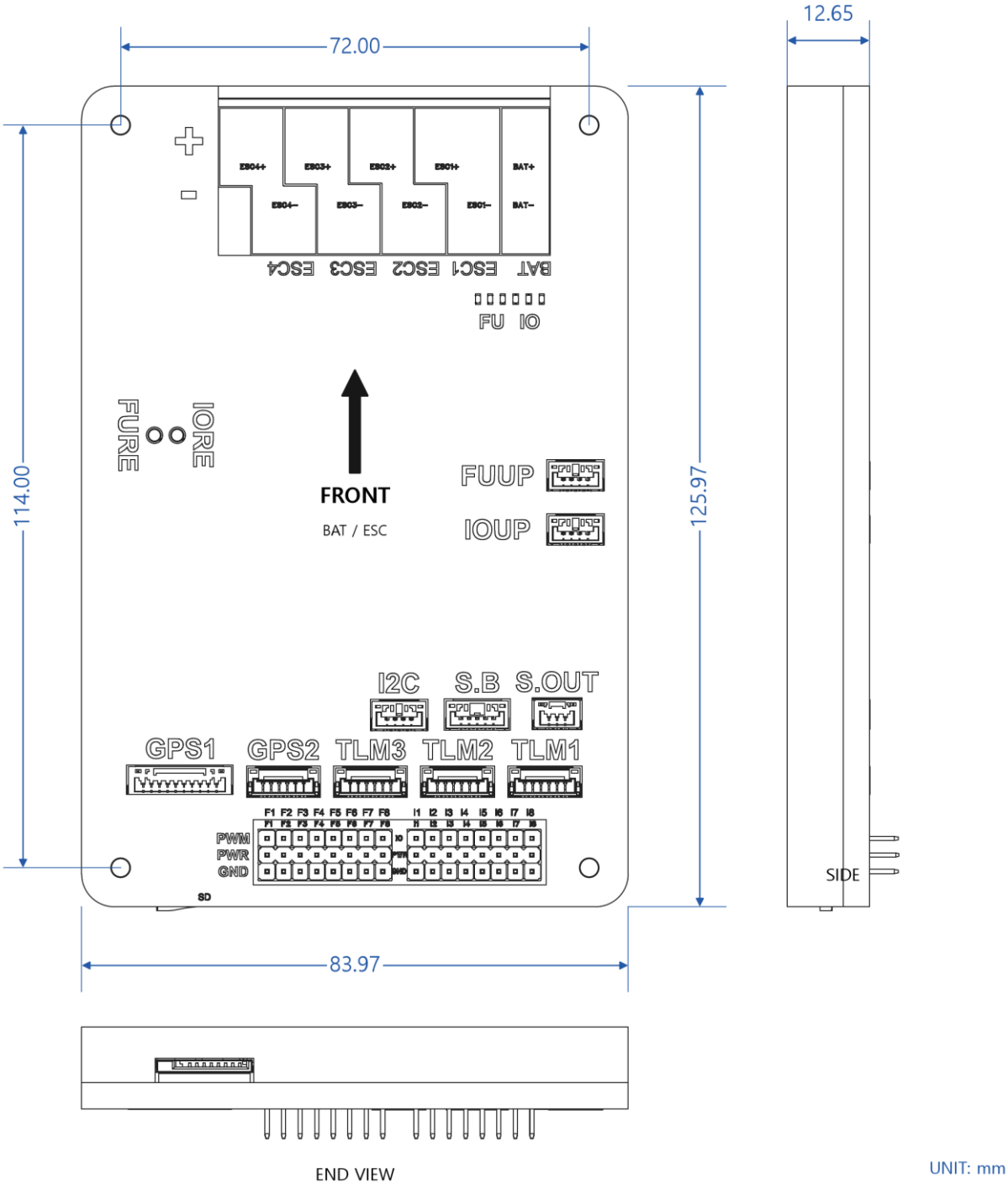
## Port power supply

| Port / power pin | Voltage | Supply current and sharing |
|---|---|---|
| **GPS1·GPS2 / TLM1~3 / I2C / S.B** | 5 V | Up to 1.5 A total, including internal FC consumption. |
| **F1–F8 + I1–I8 / PWR row** | 5 V | Up to 1.5 A total, shared by F1–F8 and I1–I8. Separate from the FC/peripheral supply. |
| **GPS1 / pin 3** | 3.3 V | 1 A supply rating, shared with microSD. |
| **S.OUT** | — | No power output pin. |
| **FUUP / IOUP / pin 1** | 3.3 V REF | Debugger voltage reference; do not use to power external devices. |
| **BAT / ESC1~4** | Battery voltage | Battery distribution to external ESCs. Up to 25.2 V (6S LiPo). |

Current figures apply to each shared supply as a whole. Allow for internal consumption and thermal limits. USB power is also subject to host and cable limits.

## Dimensions and forward direction

Front points toward the BAT and ESC power pads.



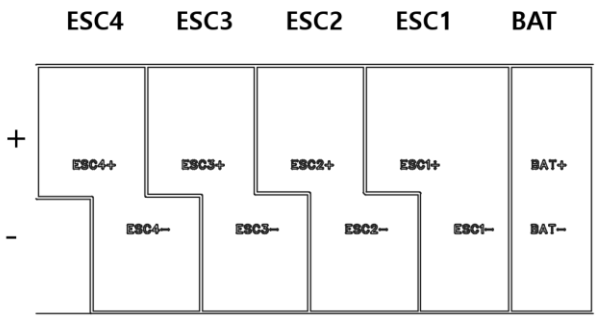

UNIT: mm

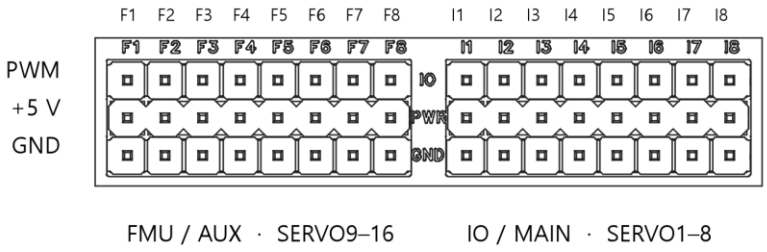
Dimensions in mm refer to the enclosure, excluding protruding connectors and pins. The 72 × 114 mm hole pattern is for enclosure assembly.

Secure the FC to the aircraft with double-sided tape. Allow clearance for USB, microSD, connectors and protruding pins.

## Power connections and PWM outputs

Top view with BAT/ESC at the top. BAT is the battery input; ESC1–4 are power pads for external ESCs.




| | |
|---|---|
| **IO PWM OUT = MAIN OUT** | I1~I8 → SERVO1~8 |
| **FMU PWM OUT = AUX OUT** | F1~F8 → SERVO9~16 |

The PWM PWR row supplies 5 V shared by IO and FMU outputs. Do not connect another BEC in parallel. Input supports 3–6S LiPo, up to 25.2 V (6S).

## Battery monitor settings

Mission Planner → CONFIG → Full Parameter List

| | | |
|---|---|---|
| **Voltage multiplier** | BATT_VOLT_MULT | 8.987575 |
| **Current scale (optional)** | BATT_AMP_PERVLT | 0.25 A/V |
| **Current offset** | BATT_AMP_OFFSET | 0 V |

BATT_MONITOR = 3 (default: voltage only) | BATT_VOLT_PIN = 8

For optional FC current monitoring, set BATT_MONITOR=4 and BATT_CURR_PIN=4, then reboot. Adjust voltage/current scale parameters against a reference meter.

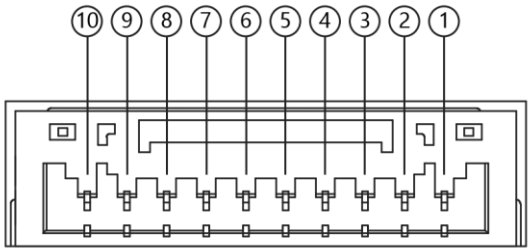
The sensor measures only the FC supply branch, excluding ESC current. Use an external sensor for total battery current, consumption and remaining capacity.

## GPS ports

Top view with Front at the top. Circled numbers identify pins. Left and right are reversed when viewed from the cable plug mating face.

## GPS1

SERIAL3 / USART1

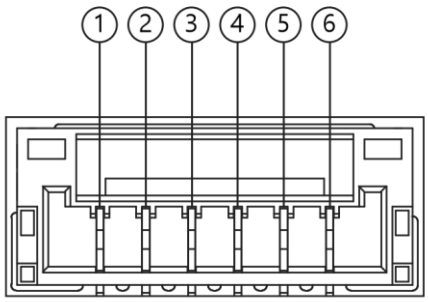



10 pins / 1.25 mm pitch

| Pin | Signal |
|---|---|
| 1 | GND |
| 2 | BUZZER |
| 3 | +3.3 V |
| 4 | SAFETY LED |
| 5 | SAFETY SW |
| 6 | SDA |
| 7 | SCL |
| 8 | RX |
| 9 | TX |
| 10 | +5 V |

## GPS2

SERIAL4 / UART8




6 pins / 1.25 mm pitch

| Pin | Signal |
|---|---|
| 1 | +5 V |
| 2 | TX |
| 3 | RX |
| 4 | SCL |
| 5 | SDA |
| 6 | GND |

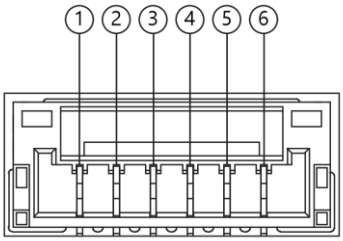
TX/RX are named from the FC perspective. Connect FC TX → device RX, FC RX ← device TX, GND ↔ GND, SCL ↔ SCL and SDA ↔ SDA. Check the peripheral power requirements.

## Telemetry and I2C ports

Top view with Front at the top. Circled numbers identify pins. Left and right are reversed when viewed from the cable plug mating face.

### TLM1

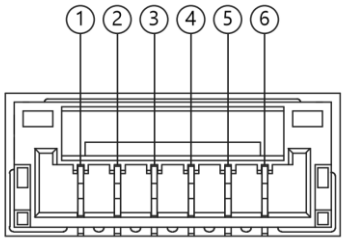
SERIAL1 / UART7




| Pin | Signal |
|---|---|
| 1 | +5 V |
| 2 | TX |
| 3 | RX |
| 4 | CTS |
| 5 | RTS |
| 6 | GND |

### TLM2

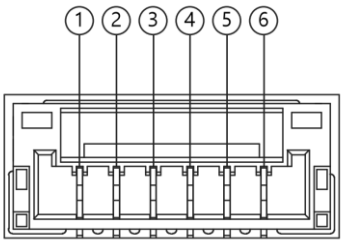
SERIAL2 / UART5




| Pin | Signal |
|---|---|
| 1 | +5 V |
| 2 | TX |
| 3 | RX |
| 4 | CTS |
| 5 | RTS |
| 6 | GND |

### TLM3

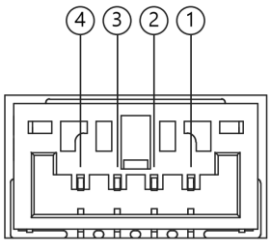
SERIAL5 / USART2




| Pin | Signal |
|---|---|
| 1 | +5 V |
| 2 | TX |
| 3 | RX |
| 4 | SCL |
| 5 | SDA |
| 6 | GND |

### I2C

I2C4 / 1.25 mm pitch




| Pin | Signal |
|---|---|
| 1 | GND |
| 2 | SDA |
| 3 | SCL |
| 4 | +5 V |

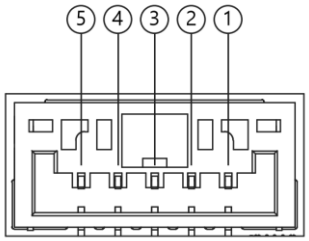
TLM1 and TLM2 have CTS/RTS. TLM3 pins 4/5 are SCL/SDA. The 5 V pin supplies peripheral power. Do not apply 5 V power to UART or I2C signal pins.

## Receiver and service ports

Top view with Front at the top. Circled numbers identify pins. Left and right are reversed when viewed from the cable plug mating face.

### S.B

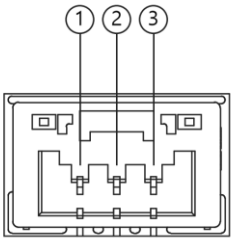
SBUS RC INPUT




| Pin | Signal |
|---|---|
| 1 | GND |
| 2 | NC |
| 3 | RSSI |
| 4 | SBUS IN |
| 5 | +5 V |

### S.OUT

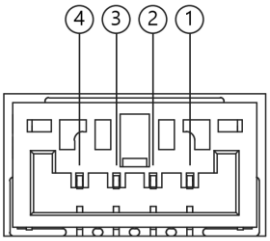
IO SBUS / OUTPUT




| Pin | Signal |
|---|---|
| 1 | NC |
| 2 | SIGNAL OUT |
| 3 | GND |

### FUUP

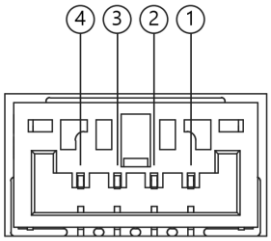
FMU SWD / SERVICE




| Pin | Signal |
|---|---|
| 1 | 3.3 V REF |
| 2 | SWCLK |
| 3 | SWDIO |
| 4 | GND |

### IOUP

IO SWD / SERVICE




| Pin | Signal |
|---|---|
| 1 | 3.3 V REF |
| 2 | SWCLK |
| 3 | SWDIO |
| 4 | GND |

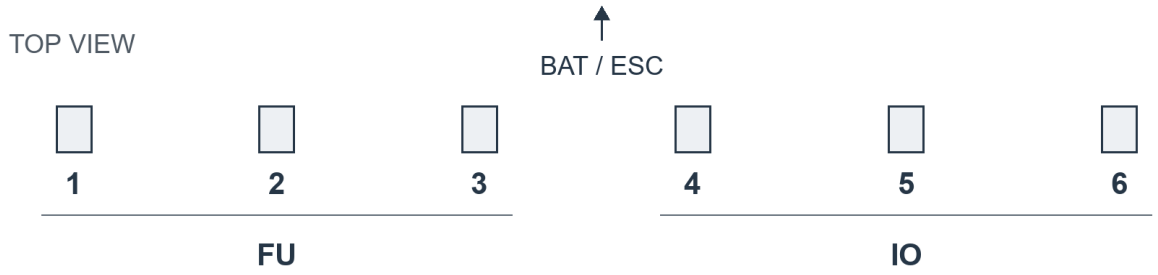
S.B is the SBUS input; S.OUT is the IO SBUS output and has no power pin. See page 10 for receiver, RSSI and SBUS output settings.

Leave NC pins unconnected. FUUP and IOUP are SWD service ports; 3.3 V REF is the debugger voltage reference.

## Reset buttons and individual LEDs

FURE resets the FMU; IORE resets the IO processor. These are not DFU buttons. Do not press them during flight.

Top view of the enclosure with BAT/ESC power pads at the top. Numbers identify the LEDs in this guide.




| Position | Function | Indication |
|---|---|---|
| 1 · FU left | FMU status | Arming, pre-arm checks and calibration. See the patterns below. |
| 2 · FU centre | GPS status | Off with no GPS or no fix. After a fix, repeated groups of flashes reflect the satellite count. |
| 3 · FU right | FMU power | On when the FMU 3.3 V supply is present. |
| 4 · IO left | FMU link warning | On if no FMU control data is received for over 200 ms. Off while data is being received normally. |
| 5 · IO centre | IO steady indicator | Held on while stock IO firmware runs. Use LED 4 to check FMU communication. |
| 6 · IO right | IO power | On when the IO 3.3 V supply is present. |

**LED 1 FMU status patterns**

| Pattern | State |
|---|---|
| Rapid flashing just after startup | Initialising |
| Slow single flashes while disarmed | Pre-arm checks passed |
| Repeated double flashes while disarmed | Pre-arm checks failed; check ground-station messages |
| Steady on | Armed, with no battery, RC or GCS failsafe |
| Flashing while armed | Battery failsafe; faster flashing for RC or GCS failsafe |

LEDs 1 and 2 alternate during trim saving or ESC calibration. Both flash briefly together during compass or IMU temperature calibration.

These are the normal indications while ArduPilot runs. Bootloader, firmware-update and AutoTune notifications use other patterns. Power LEDs alone do not confirm correct operation.

## UART and USB

| Parameter | Connection | Default protocol |
|---|---|---|
| SERIAL0 | First USB virtual port | MAVLink |
| SERIAL1 | TLM1 / UART7 / CTS/RTS | MAVLink2 |
| SERIAL2 | TLM2 / UART5 / CTS/RTS | MAVLink2 |
| SERIAL3 | GPS1 / USART1 | GPS |
| SERIAL4 | GPS2 / UART8 | GPS |
| SERIAL5 | TLM3 / USART2 | None |
| SERIAL6 | USART3 / not pinned out | None |
| SERIAL7 | Second virtual port on the same USB | MAVLink2 |

All UARTs have RX and TX DMA. USART6 is the internal link between the FMU and IO processors.

**Loading firmware**

The board ships with an ArduPilot-compatible bootloader. Connect USB and upload a DCD-FC .apj file using Mission Planner or another compatible ground station. Normal updates do not require DFU mode or ST-Link.

Saved parameters are retained after firmware updates. Retain a working stock IO bootloader.

After board support is merged and builds are published, use the DCD-FC target at firmware.ardupilot.org.

## Outputs and receiver settings

### PWM and DShot

All 16 outputs support PWM and DShot. To use Bi-Directional DShot on I1-I4, set BRD_IO_DSHOT=1 and reboot.

| Board outputs | ArduPilot outputs | Timer group |
|---|---|---|
| I1-I2 | SERVO1-2 | IO TIM2 |
| I3-I4 | SERVO3-4 | IO TIM4 |
| I5-I8 | SERVO5-8 | IO TIM3 |
| F1-F4 | SERVO9-12 | FMU TIM1 |
| F5-F6 | SERVO13-14 | FMU TIM4 |
| F7-F8 | SERVO15-16 | FMU TIM5 |

Channels within the same group must use the same output rate. If any channel uses DShot, all channels in that group must use DShot.

### Receiver and RSSI

Connect SBUS to S.B pin 4. PPM is not supported. Connect a CRSF/ELRS receiver to the TX/RX pins of TLM3 and set SERIAL5_PROTOCOL=23.

For analog RSSI on S.B pin 3, set RSSI_TYPE=1 and RSSI_ANA_PIN=103. S.OUT is the IO SBUS output, enabled with BRD_SBUS_OUT. SBUS output and analog RSSI cannot be used simultaneously.

### GPIO

| Board outputs | GPIO numbers |
|---|---|
| F1-F8 | 50-57 |
| I1-I8 | 101-108 |

Use these numbers when configuring output pins as GPIOs. GPIO pins carry signals; they are not power supplies.

### Compass

An IST8310 compass is fitted. Due to potential magnetic interference, an external I2C compass in a GPS/compass module is recommended.